GlobalGeoTree Germany, Berlin and NRW visualisation report

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0.1 Purpose

This is one cleaned and unified QMD for the Germany, Berlin, and Nordrhein-Westfalen (NRW) parts of the GlobalGeoTree analysis. The rendered HTML hides the code (`echo: false`) so the report reads like a normal analysis report. The code in the QMD source is still commented and readable.

The dataset used here is always called `GlobalGeoTree.csv`.

0.2 What I changed after checking the older QMDs and PDFs

Old visualisation / section	Decision	Reason
Germany overview table	Keep	It gives the two most important checks before any maps: the number of German observations and the number of recorded species.
Germany distribution map	Improve	The old grid map was too pale and hard to interpret. The default map is now a smoother hotspot map, and two alternative map types are included in the setup.
Observations by year	Keep	It shows whether the German subset is dominated by a few recent years. This is important context for all later maps.
Observations by source: Germany map and bar chart	Keep	The map shows where the main sources are active. The bar chart shows which source groups dominate the dataset.
Observations mapped to Bundesländer, plus Bundesländer table, bar chart and map	Keep	The table checks the spatial join, the bar chart ranks states, and the map shows the regional pattern.
Berlin district boundary loading and Berlin observations table	Keep	These are essential checks before any Berlin district or density map.

Old visualisation / section	Decision	Reason
Berlin observation/species bar chart	Remove	Observations and species are different measures on very different scales, so the table is clearer than a two-bar plot.
Berlin source table and source bar chart	Keep	Berlin is strongly shaped by a few sources, so source composition should be shown.
Spatial distribution of Berlin observations	Keep	This was the strongest Berlin visual because it clearly shows observation hotspots inside the city.
Berlin local-area table, bar chart, species chart and map	Keep, but rename carefully	The downloaded Berlin file has about 97 polygons, so these are better described as Berlin local areas / district polygons rather than the 12 official Bezirke.
Berlin source-by-district heatmap	Remove from the main report	With many local areas and long source names, it becomes visually crowded and hard to read.
Berlin top recorded species table and bar chart	Keep	It adds useful biological interpretation, as long as the caption says it is based on recorded observations, not true abundance.
NRW boundary check and summary table	Add	The NRW PDF already had a useful boundary workflow and summary, so it belongs in this unified report.
NRW source table and source bar chart	Add	NRW is strongly dominated by a few sources, so the source chart is needed.
NRW observation map	Add, but improve	A raw point map can overplot, so the default NRW map uses a density/hotspot layer with the NRW boundary.
NRW district table, district bar chart, species bar chart and district map	Add	These are useful for comparing NUTS3 districts / Kreise and kreisfreie Städte.

Old visualisation / section	Decision	Reason
NRW source map	Add as a faceted source map	It helps check whether the largest sources have different spatial footprints.
NRW common-name / GBIF enrichment	Remove	It depends on internet access and extra packages. Scientific names are more reproducible.
NRW category composition	Remove	The user confirmed that category should be ignored because it is not available in the full data file.
Germany category and climate-zone sections	Remove	Category is not available in the full file, and the climate-zone section needs an extra polygon file.
Very wide first-10-row preview tables	Remove	They were not readable in the rendered report and did not directly answer the visualisation questions.

0.3 Density map options used in this report

The report now keeps several density-style maps so you can compare them directly in the rendered HTML. This is better than forcing one single map style too early.

The main comparison styles are:

1. **Points + contours**: good when you want to see both the individual recorded points and the broader hotspot shapes.
2. **Smooth filled density**: good for a clean presentation view of hotspots.
3. **Gridded heatmap**: good when you want counts in visible cells rather than a smoothed surface.
4. **Sampled point map**: good as an honest reference map showing raw record locations.

My recommendation is to keep **points + contours** and **smooth filled density** in the final report, and use the **gridded heatmap** as a comparison/checking visual.

0.4 Setup

0.5 Load and clean the observation data

The data file is large, so the script reads only the columns needed for this report. This keeps the report faster and makes the workflow easier to understand.

Input data check

item	value
Input file	GlobalGeoTree.csv
Rows after coordinate/species cleaning	6,263,345
Countries after cleaning	221
Germany rows used in this report	205,747

0.6 Germany overview table

Why this table is needed: before drawing any maps, I want a quick check of how much German data is available. This also keeps the requested metric table in the report.

Germany overview

metric	value
Germany observations	205,747
Germany recorded species	464

Interpretation. The observations value counts records, while the recorded species value counts unique scientific species names. These two numbers answer different questions, so they should stay separate.

0.7 Load Bundesländer boundaries

The Germany state maps need Bundesländer polygons. The code first looks for a local GeoJSON file and then tries the public GISCO NUTS1 file.

Bundesländer boundary check

item	value
Boundary status	Loaded Bundesländer / NUTS1 boundaries from data/external/germany_nuts1_2021.geojson
Number of Bundesländer polygons	16

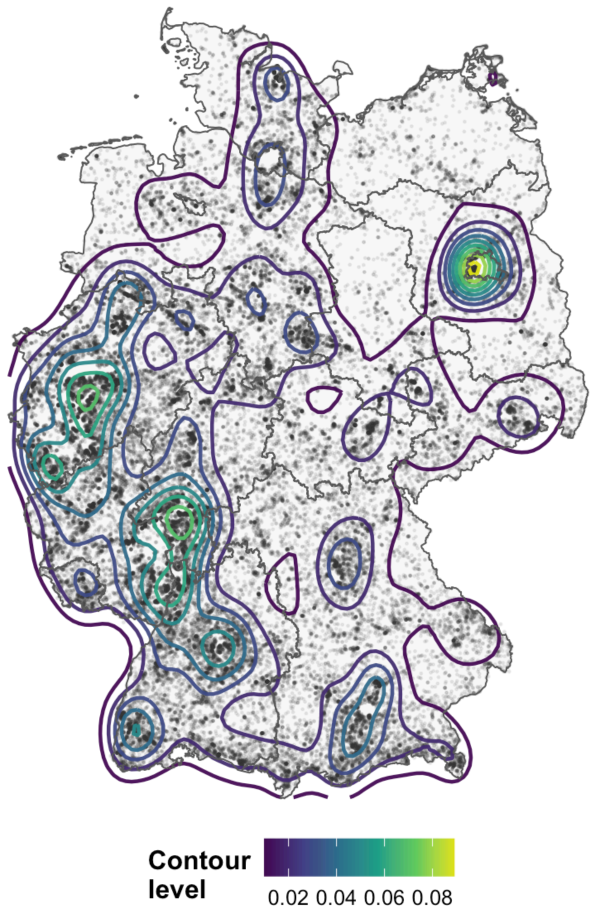
0.8 Distribution of observations in Germany

Why this visual is needed: the dataset contains coordinates, so a map is the fastest way to see where records are concentrated. Because one map type does not tell the whole story, the report now keeps several Germany density visualisations for comparison.

Suggested reading order: start with the **points + contours** map, check the **smooth filled density** map for a cleaner hotspot view, and use the **grid map** as a non-smoothed count-based comparison.

Germany observations: points plus density contours

Contours show broader hotspots, while semi-transparent points keep the underlying rec



Displayed points: 70,000 of 205,747 Germany observations.

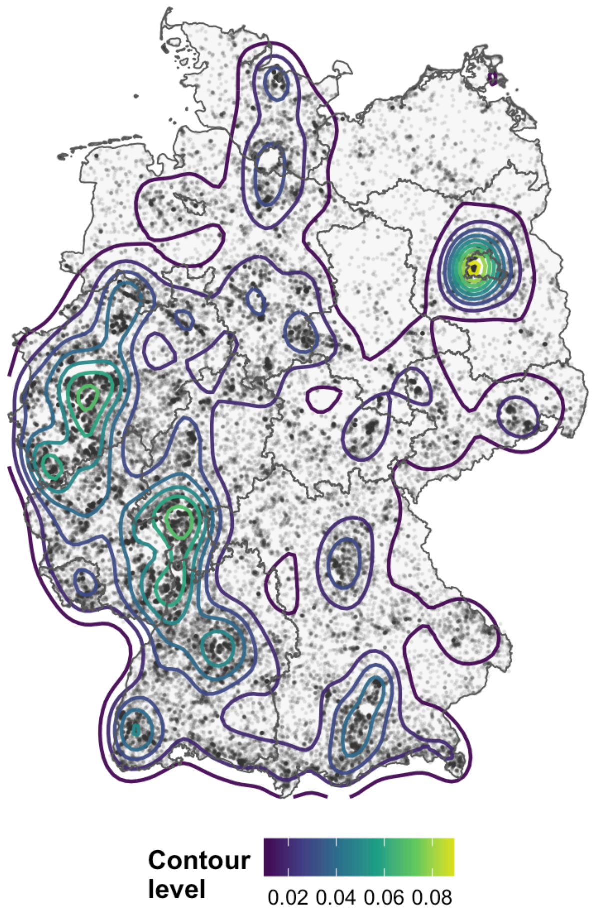
Interpretation. Darker/higher-density areas show places with more recorded observations. This should be interpreted as observation intensity, because a place can have many records due to active platforms, active observers, or easier access.

0.8.1 Germany density map comparison

These extra maps are kept on purpose so you can compare styles in the rendered HTML.

Germany observations: points plus density contours

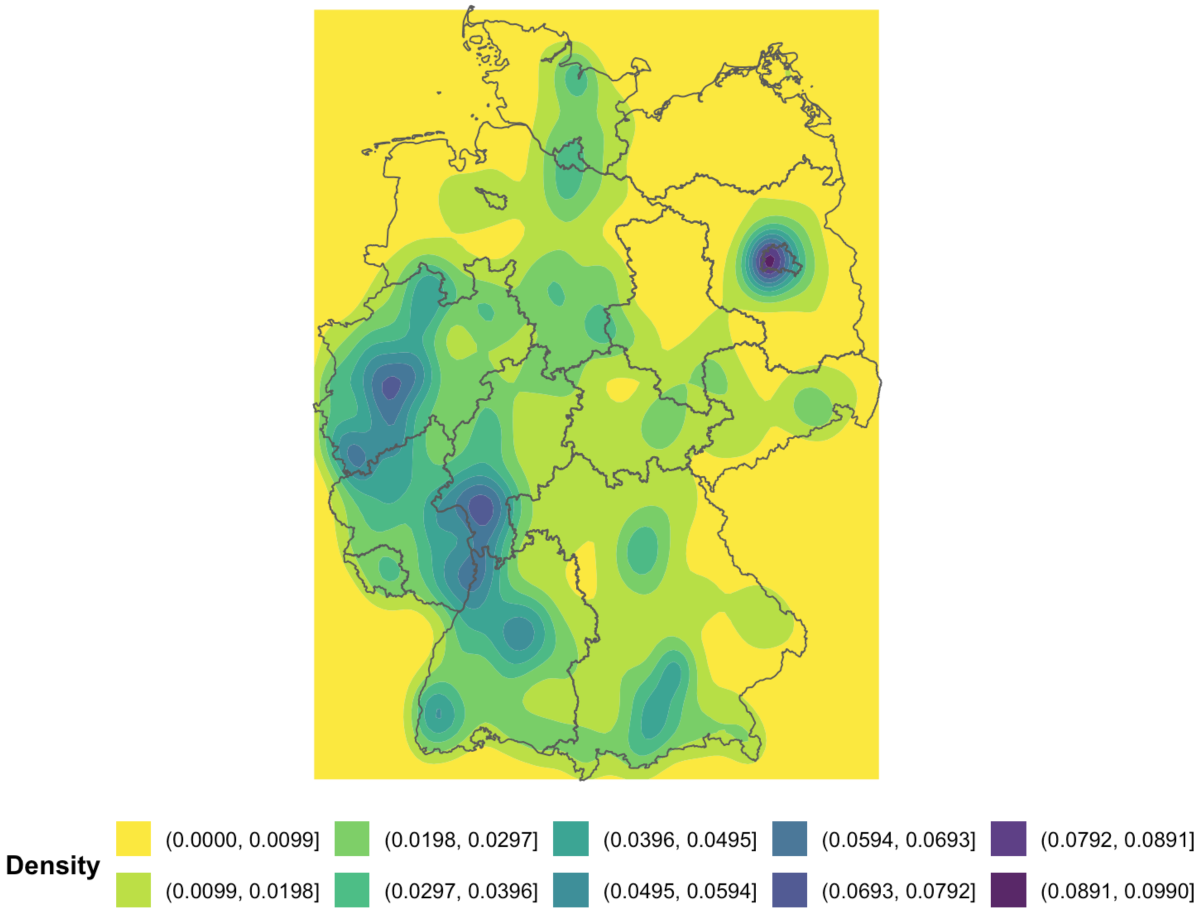
Best balanced option: visible points plus coloured hotspot contours.



Good compromise between raw records and broad hotspot structure.

Germany observations: smooth filled density

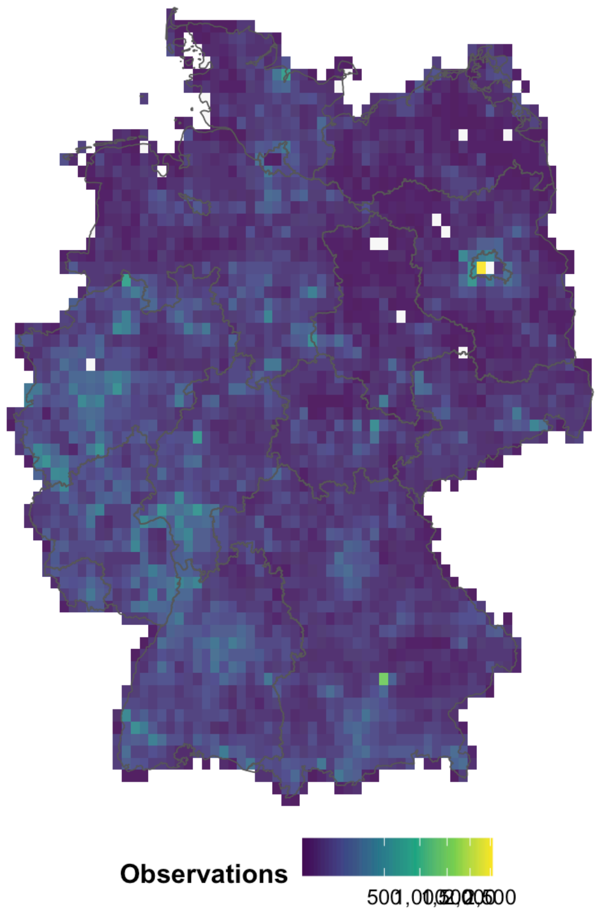
Clean presentation option, but smoothing can make sparse areas look more continuou



Best for a clean visual summary of hotspots.

Germany observations: gridded heatmap

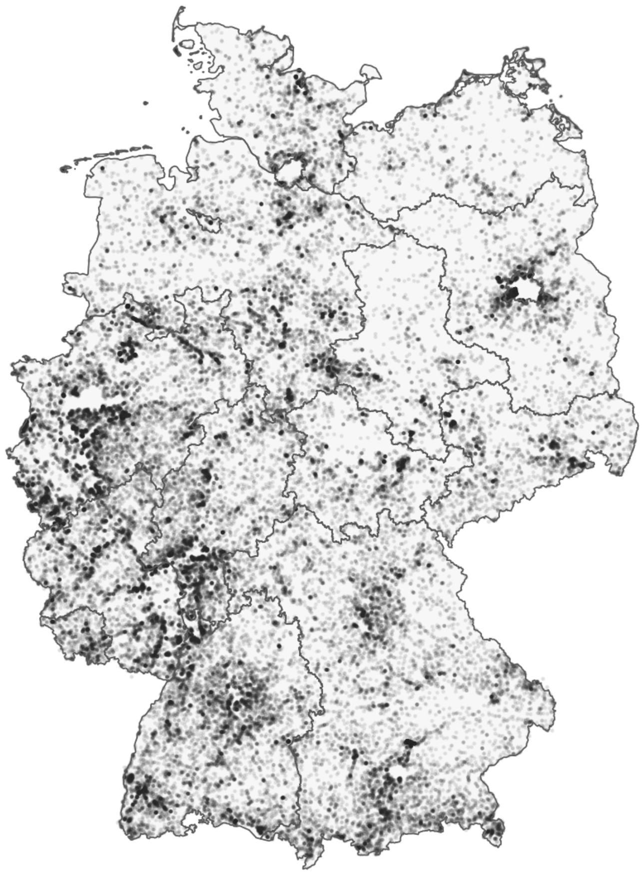
Each square counts observations directly, so this view is less smoothed than the densit



Useful when you want visible count cells instead of a smoothed surface.

Germany observations: sampled points

Raw-point reference map with no smoothing.

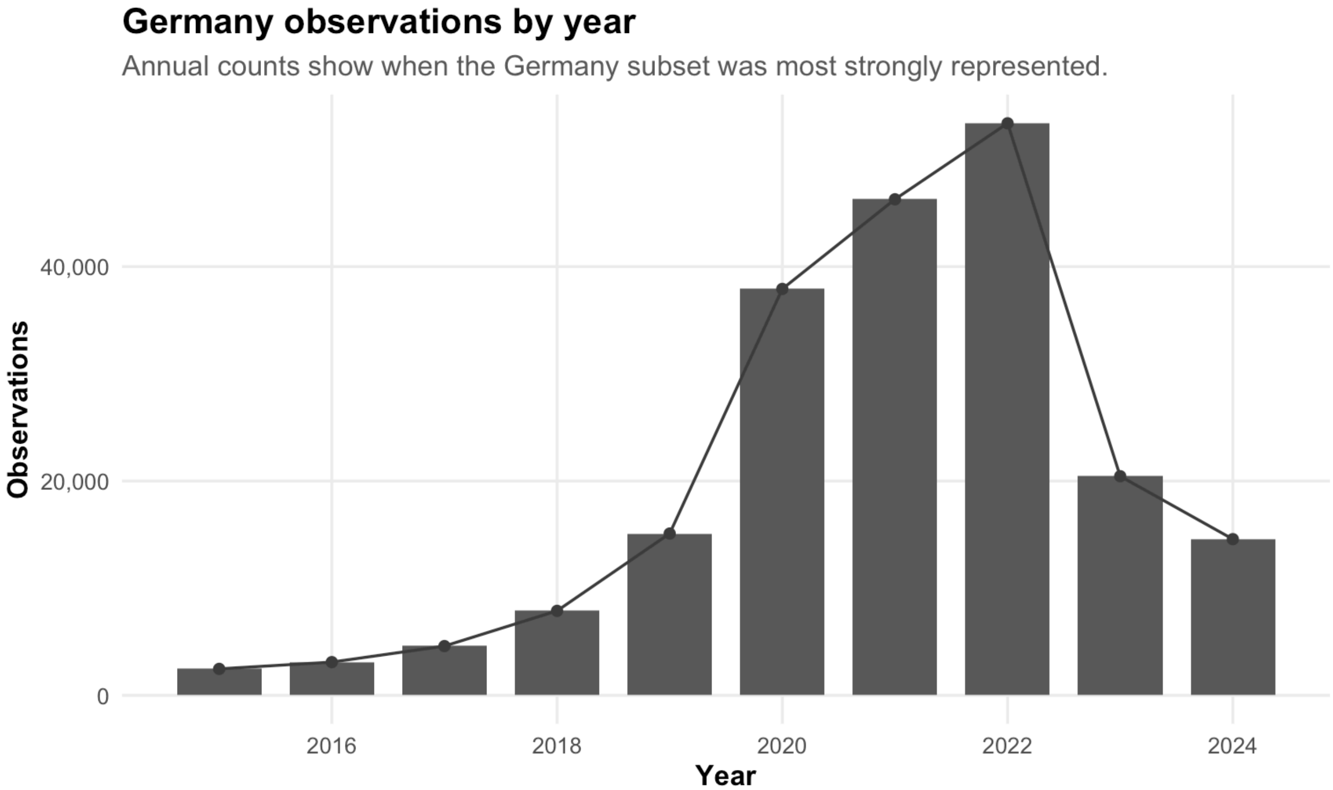


Displayed points: 70,000 of 205,747 Germany observations.

Quick recommendation. If you want one final Germany map, I would keep **points + contours** as the main option and **smooth filled density** as the backup presentation option.

0.9 Observations by year

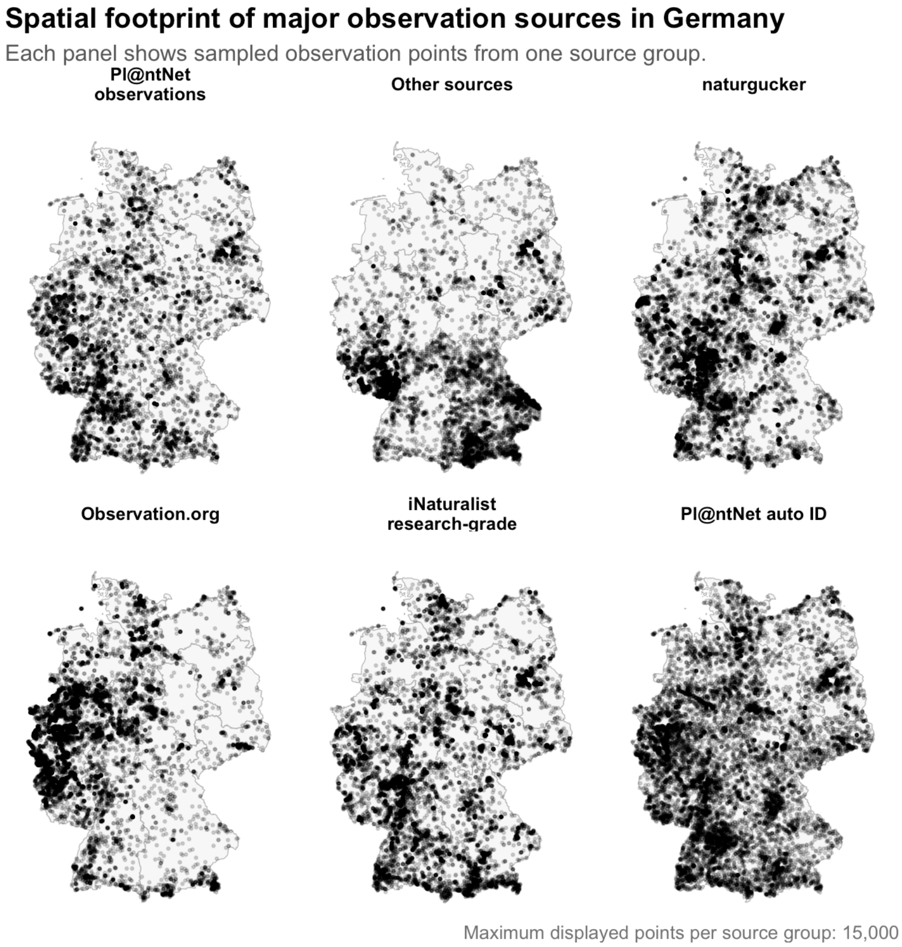
Why this visual is needed: the year chart checks whether the dataset is balanced through time or dominated by a few recent years. This matters because spatial maps can be influenced by when observations were collected.



Interpretation. Peaks in this chart should be read as peaks in recording activity. They do not automatically mean that more trees existed in those years.

0.10 Observations by source on the country map

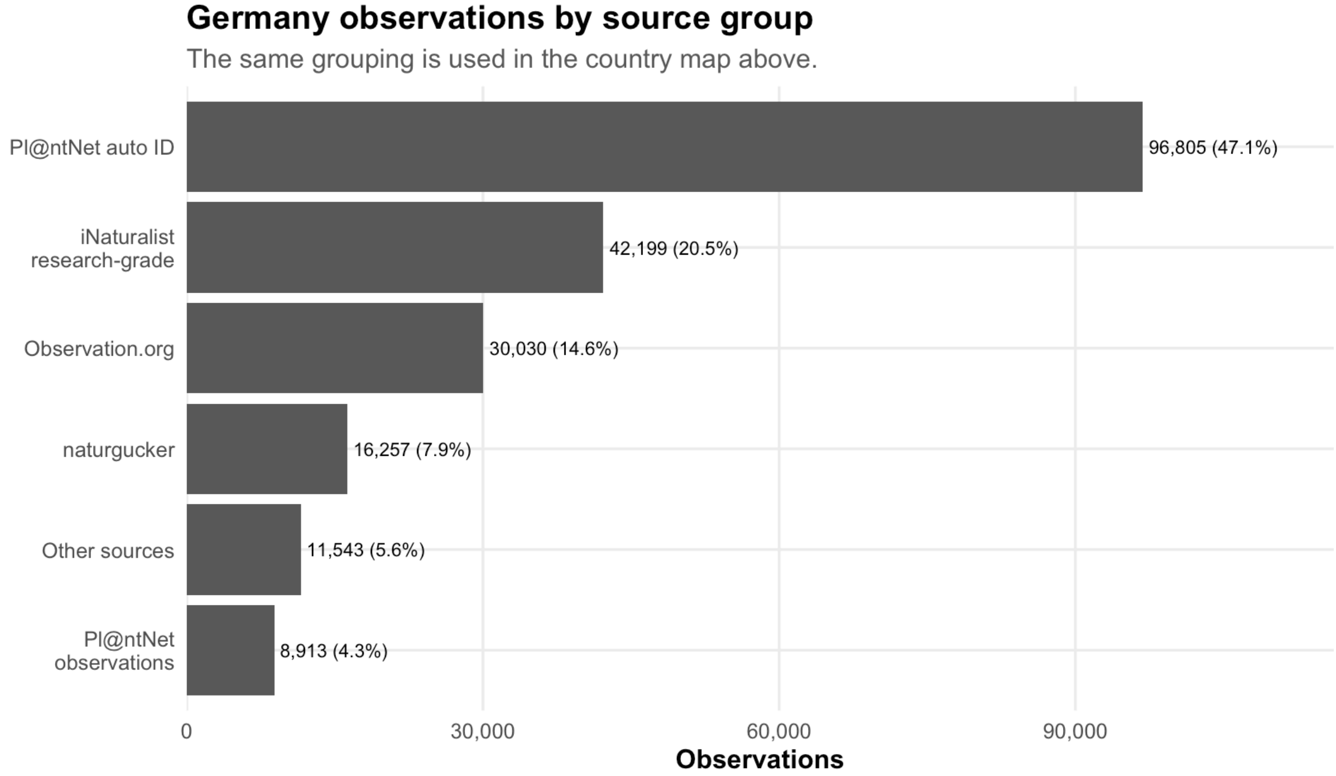
Why this visual is needed: different data sources can have different spatial footprints. The map helps check whether one source dominates certain areas of Germany.



Interpretation. This map is useful for spotting source bias. If one source is dense in one region and another source is dense elsewhere, regional patterns may partly reflect data collection rather than biology.

0.11 Observations by source as a bar chart

Why this visual is needed: the map shows geography, but the bar chart makes it easier to compare source sizes. Both are needed because they answer different questions.



Interpretation. The largest bar shows which source contributes the most records. This is important because a dominant source can strongly shape the spatial pattern seen on the maps.

0.12 Short note on the two Pl@ntNet source names

`Pl@ntNet observations` and `Pl@ntNet auto ID` are both Pl@ntNet-derived, but they do not mean exactly the same thing. In this report, I keep them separate because they use different validation routes: one is the regular shared-observation dataset, and the other is the automatically identified occurrences dataset.

0.13 Observations mapped to Bundesländer

Why this step is needed: the original data has coordinates, not a ready-made Bundesland field. To compare federal states, the points must first be spatially joined to the Bundesländer polygons.

Bundesland spatial join check

metric	value
Germany observations before Bundesland join	205,747
Germany observations matched to a Bundesland	205,143
Bundesländer with at least one observation	16

Observations mapped to Bundesländer

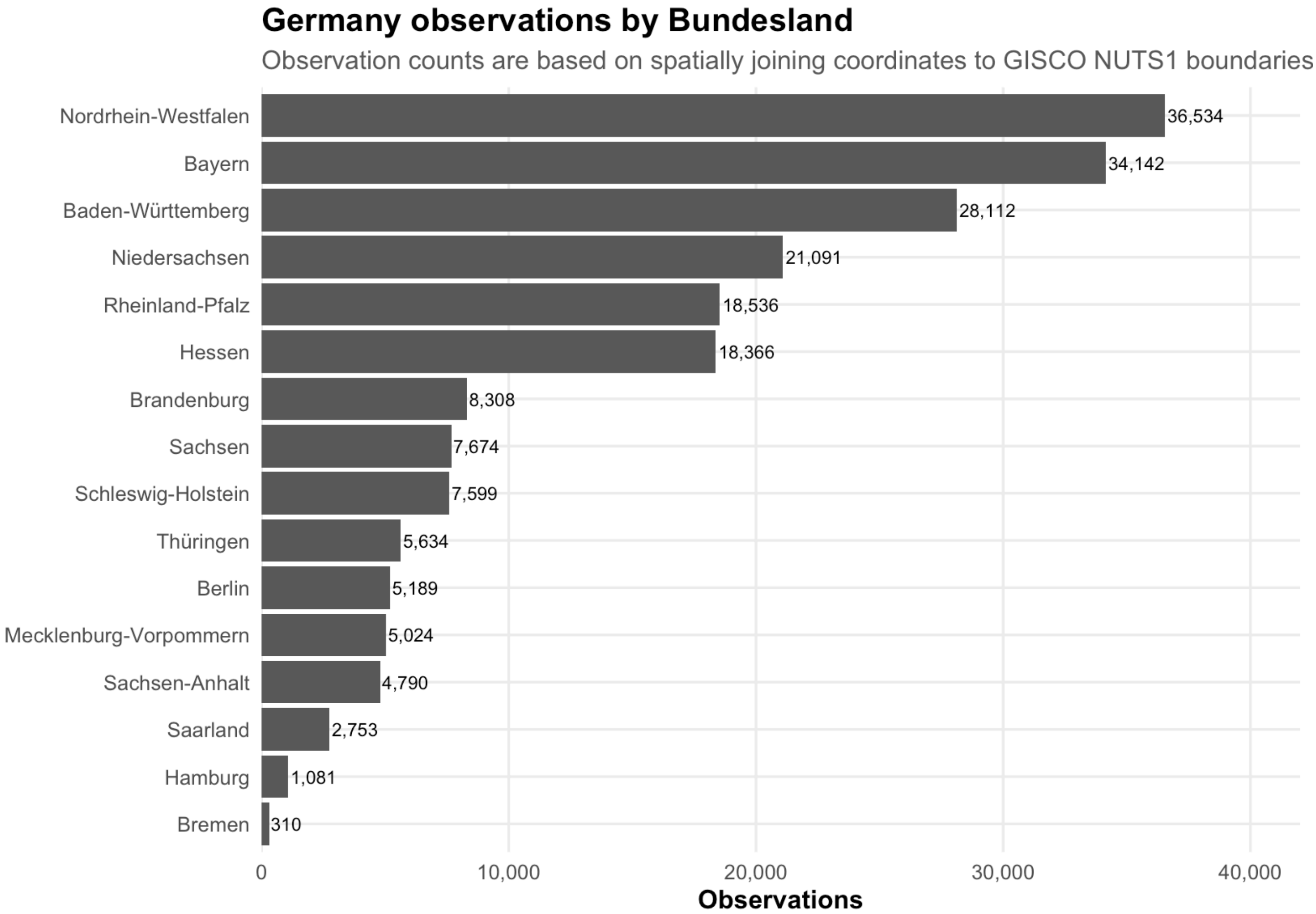
bundesland	observations	species	sources
Nordrhein-Westfalen	36,534	243	16
Bayern	34,142	301	25

bundesland	observations	species	sources
Baden-Württemberg	28,112	287	21
Niedersachsen	21,091	228	16
Rheinland-Pfalz	18,536	235	17
Hessen	18,366	237	13
Brandenburg	8,308	196	14
Sachsen	7,674	196	14
Schleswig-Holstein	7,599	175	11
Thüringen	5,634	168	12
Berlin	5,189	158	13
Mecklenburg-Vorpommern	5,024	185	11
Sachsen-Anhalt	4,790	165	8
Saarland	2,753	141	11
Hamburg	1,081	106	8
Bremen	310	68	6

Interpretation. This table is the cleanest check that the spatial join worked. If many Germany points were not matched, the boundary data or coordinate quality should be checked before trusting the maps.

0.14 Bundesländer bar chart

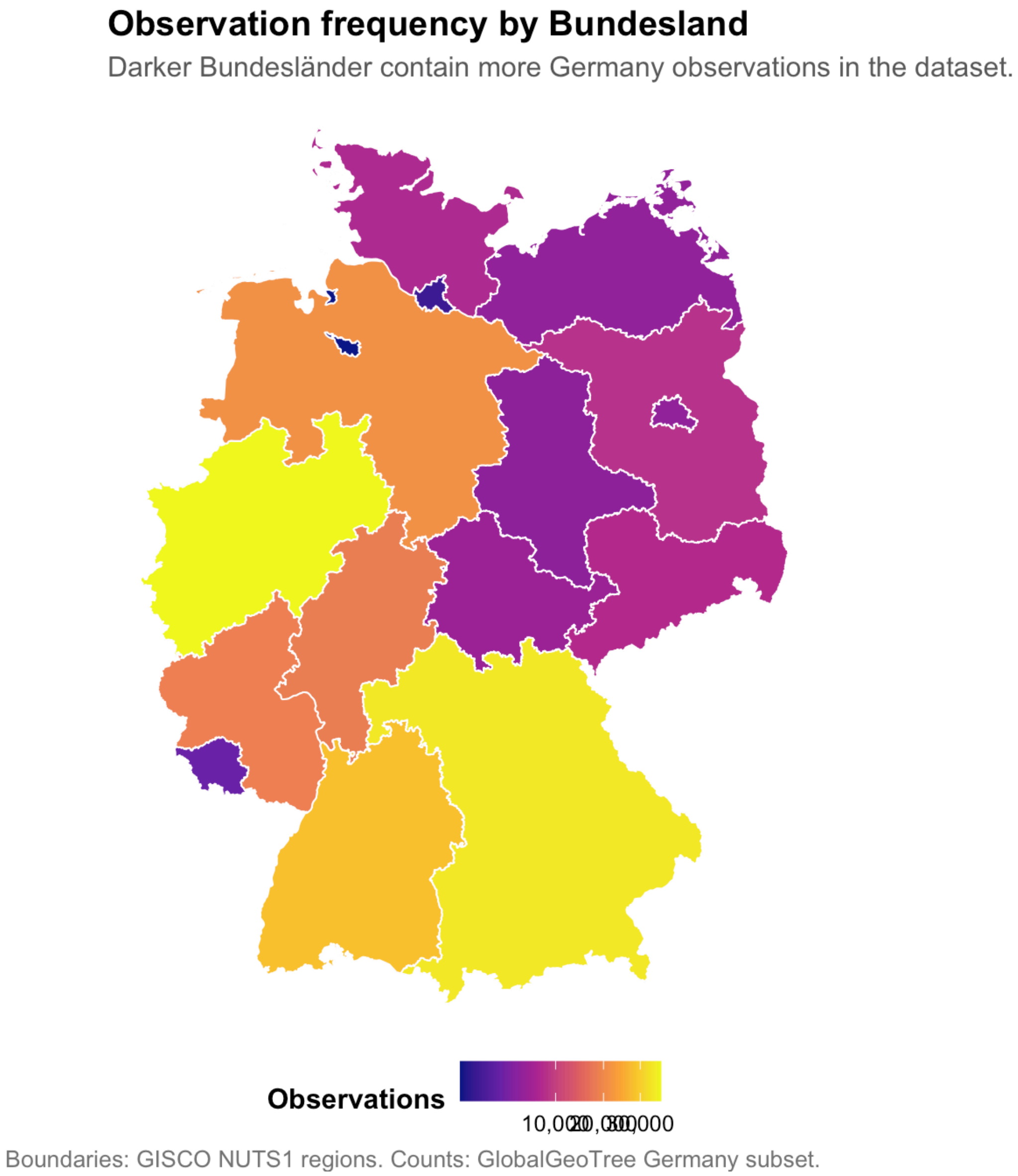
Why this visual is needed: the bar chart is the best way to rank federal states because each state is compared on the same axis.



Interpretation. This chart shows which Bundesländer dominate the German subset. It should not be read as a direct comparison of tree abundance because the number of observations also depends on recording effort.

0.15 Bundesländer map

Why this visual is needed: the map complements the bar chart by showing where high-record states are located geographically.



Interpretation. This map is useful for spatial context, but it is less precise for ranking than the bar chart. Larger states can also look visually more important because they occupy more map area.

1 Berlin

1.1 Load Berlin district boundaries

Berlin is handled separately because it needs a finer boundary layer than Bundesländer. The public fallback file contains many small Berlin local-area polygons, so I describe the results as Berlin district/local-area results.

Berlin boundary check

item	value
Berlin district/local-area boundary status	Loaded Berlin district/local-area boundaries from data/external/berlin_districts.geojson
Number of Berlin polygons	97

1.2 Berlin observations table

Why this table is needed: it shows how Berlin observations were selected and gives the main record counts before the Berlin maps.

Berlin observation summary

metric	value
Berlin filtering method	Spatial join to Berlin district/local-area polygons
Berlin observations	5,192
Berlin recorded species	158
Available source values	13
Available years	2015-2024

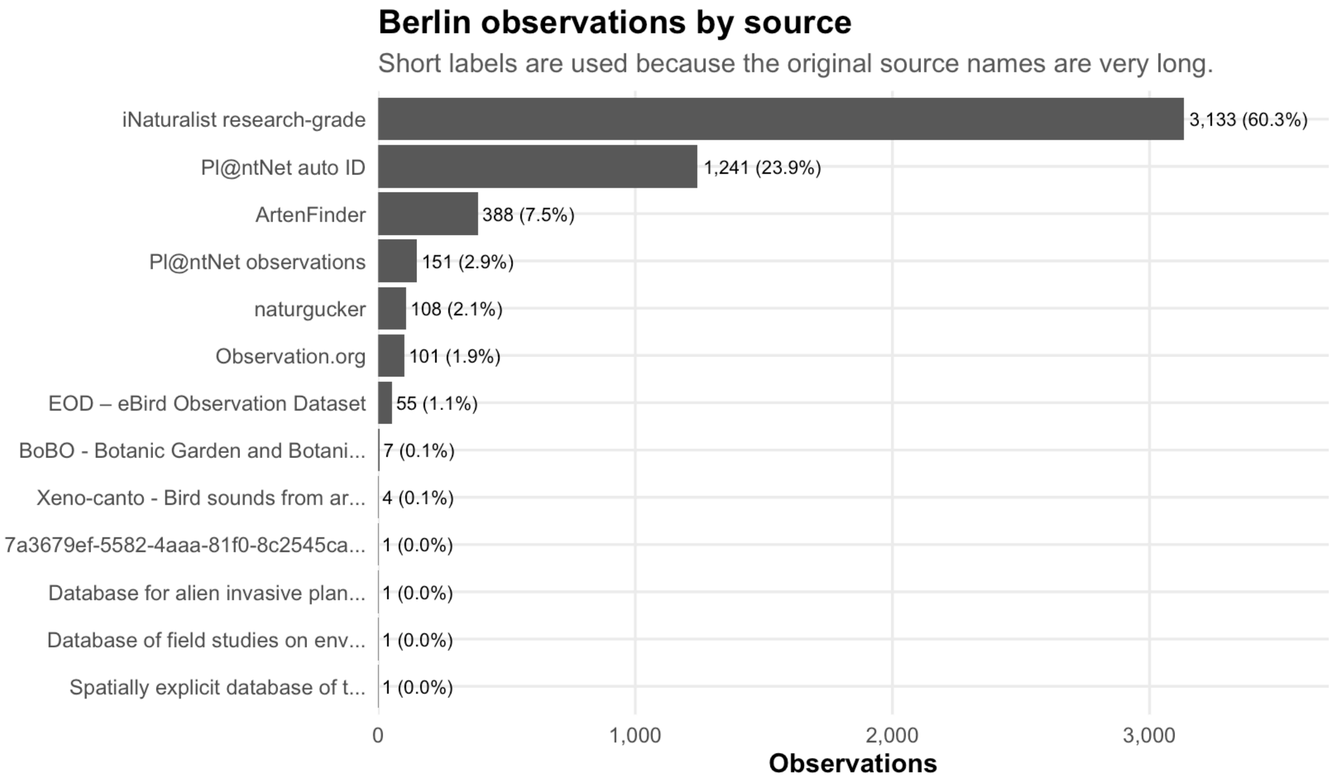
Interpretation. I removed the old observation/species bar chart because it put two very different measures on one axis. This table is clearer.

1.3 Berlin observation sources

Why this visual is needed: Berlin observations are not equally distributed across sources, so source composition is important context for every Berlin map.

Berlin observations by source

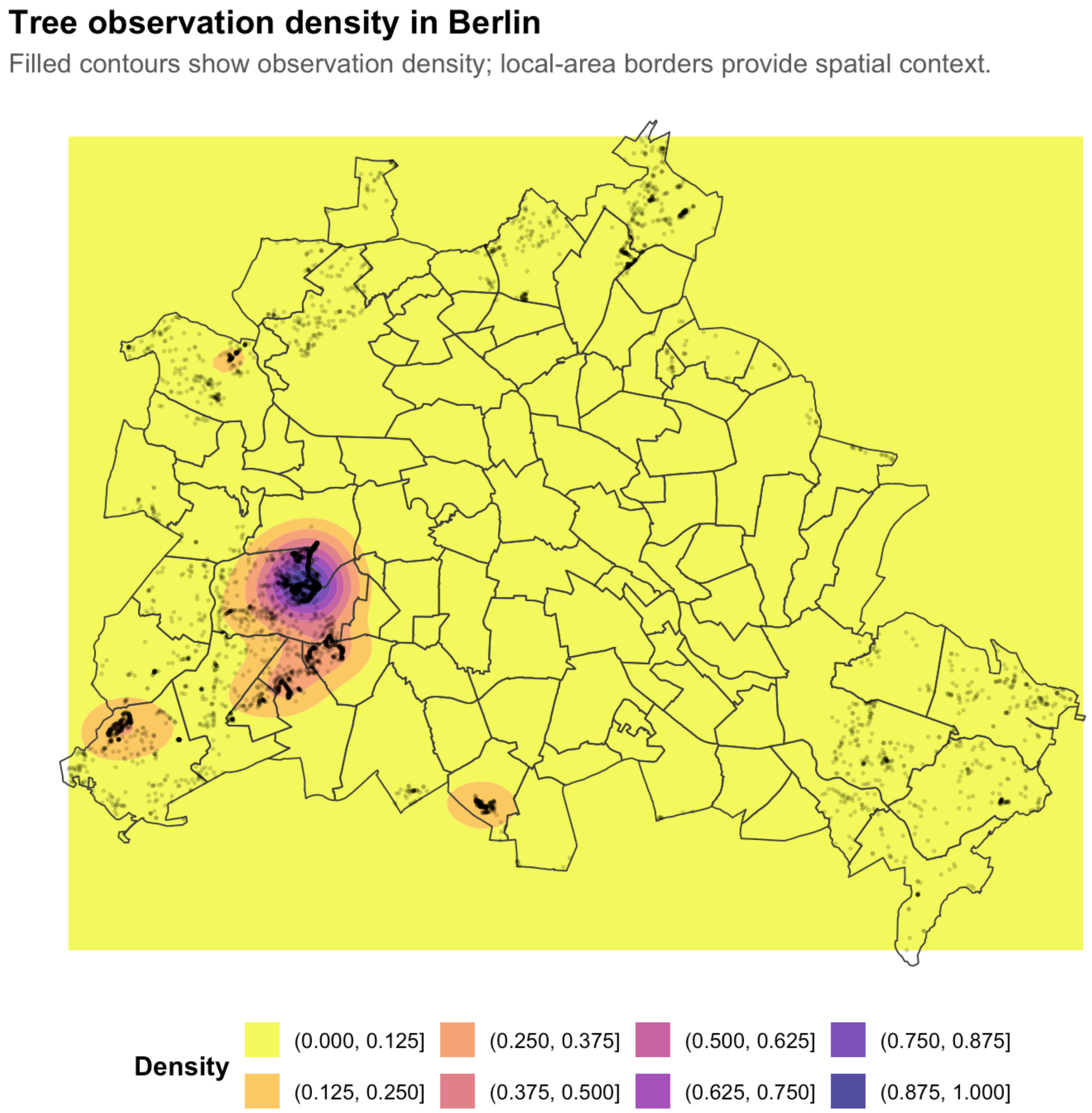
source_short	observations	share	label
iNaturalist research-grade	3,133	60.3%	3,133 (60.3%)
Pl@ntNet auto ID	1,241	23.9%	1,241 (23.9%)
ArtenFinder	388	7.5%	388 (7.5%)
Pl@ntNet observations	151	2.9%	151 (2.9%)
naturgucker	108	2.1%	108 (2.1%)
Observation.org	101	1.9%	101 (1.9%)
EOD – eBird Observation Dataset	55	1.1%	55 (1.1%)
BoBO - Botanic Garden and Botani…	7	0.1%	7 (0.1%)
Xeno-canto - Bird sounds from ar…	4	0.1%	4 (0.1%)
7a3679ef-5582-4aaa-81f0-8c2545ca…	1	0.0%	1 (0.0%)
Database for alien invasive plan…	1	0.0%	1 (0.0%)
Database of field studies on env…	1	0.0%	1 (0.0%)
Spatially explicit database of t…	1	0.0%	1 (0.0%)



Interpretation. This chart tells us which platforms shape the Berlin subset most. If one source dominates, the Berlin hotspots may partly show that source's recording behaviour.

1.4 Spatial distribution of Berlin observations

Why this visual is needed: this is the clearest Berlin plot because it shows where observations cluster inside the city. The density layer avoids overplotting, and the local-area borders give spatial context.



Interpretation. The density map highlights observation hotspots inside Berlin. Like the Germany maps, it is a map of recorded observations, not a direct map of all trees in the city.

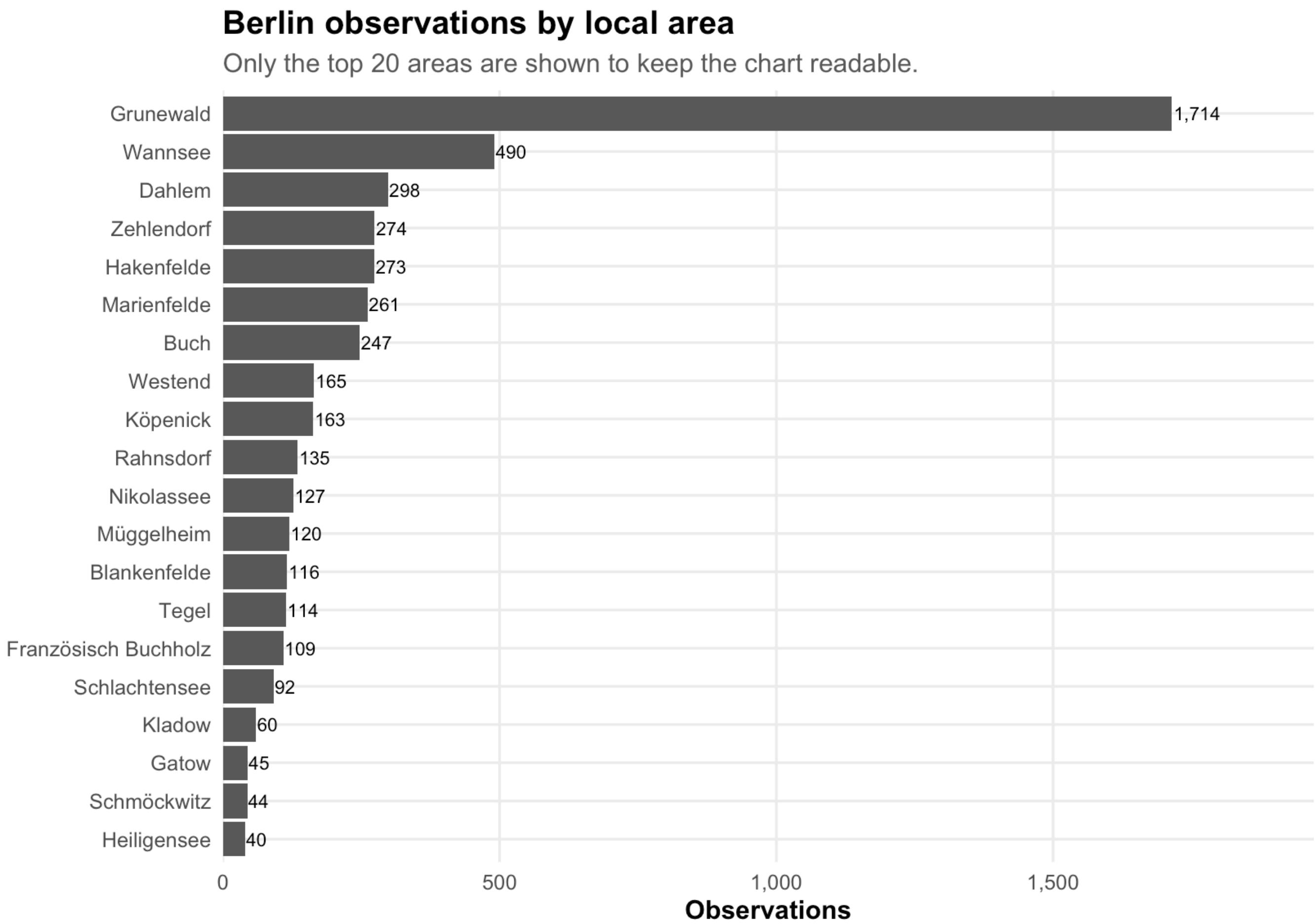
1.5 Berlin local areas: observations and species

Why this section is needed: the density map shows hotspots, but district/local-area summaries make the pattern easier to compare and check.

Top Berlin local areas by observation count

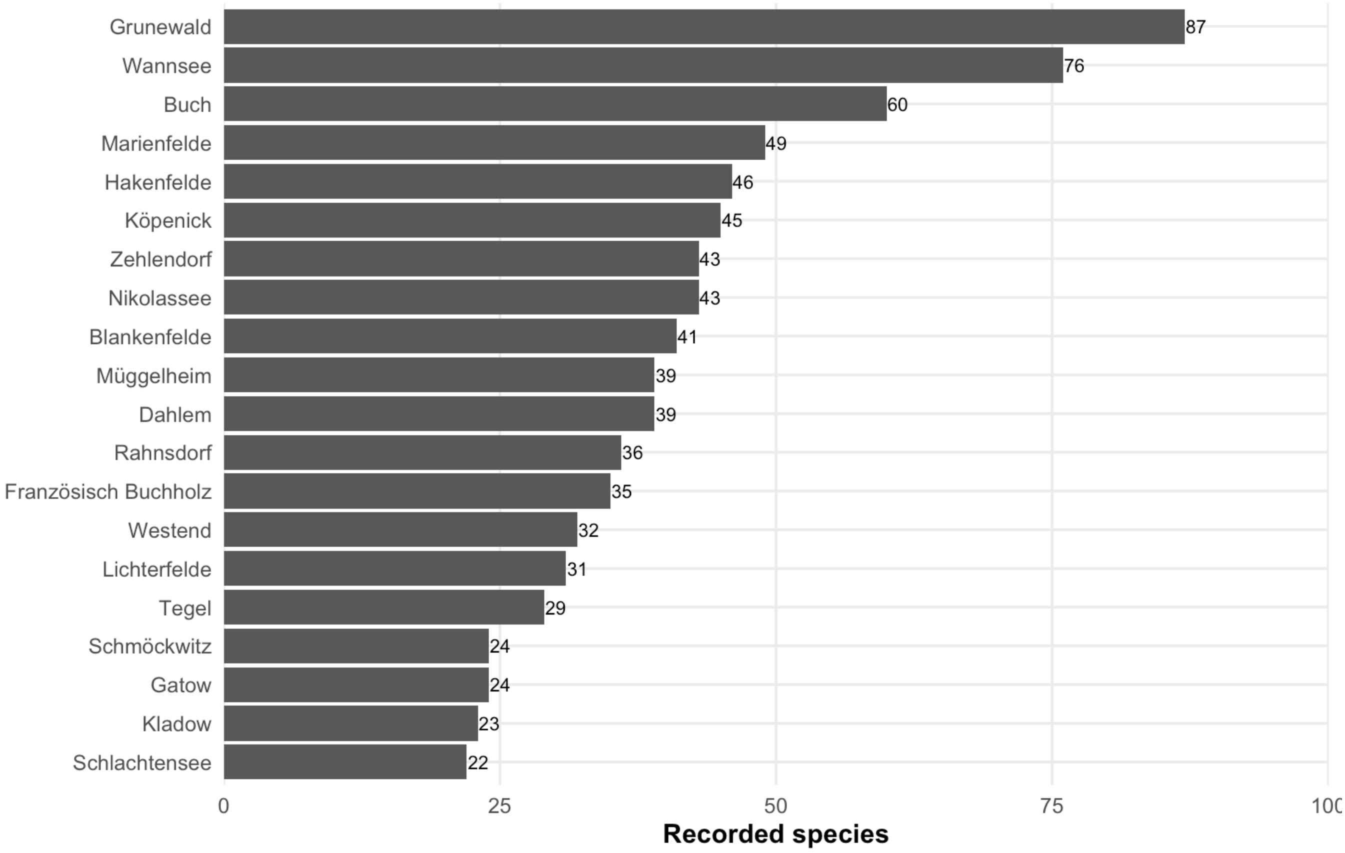
district	observations	species	sources
Grunewald	1,714	87	7
Wannsee	490	76	8
Dahlem	298	39	7
Zehlendorf	274	43	6
Hakenfelde	273	46	8
Marienfelde	261	49	6

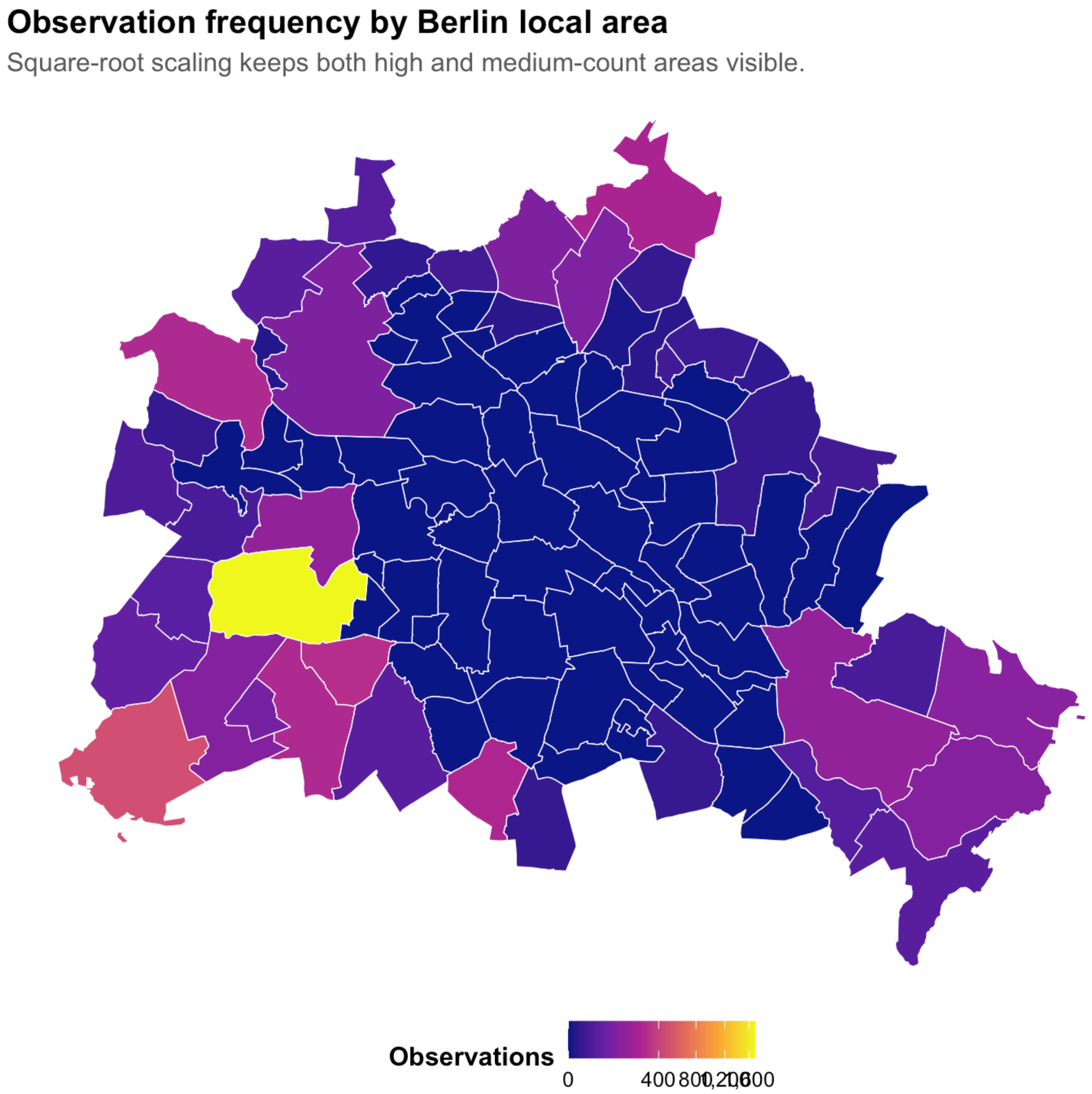
district	observations	species	sources
Buch	247	60	7
Westend	165	32	5
Köpenick	163	45	7
Rahnsdorf	135	36	5
Nikolassee	127	43	7
Müggelheim	120	39	8
Blankenfelde	116	41	7
Tegel	114	29	8
Französisch Buchholz	109	35	5
Schlachtensee	92	22	4
Kladow	60	23	5
Gatow	45	24	5
Schmöckwitz	44	24	7
Heiligensee	40	15	5



Recorded species by Berlin local area

This counts distinct scientific species names recorded in each area.





Interpretation. The bar charts are better for comparing exact ranks, while the map is better for seeing where the high-record areas are located.

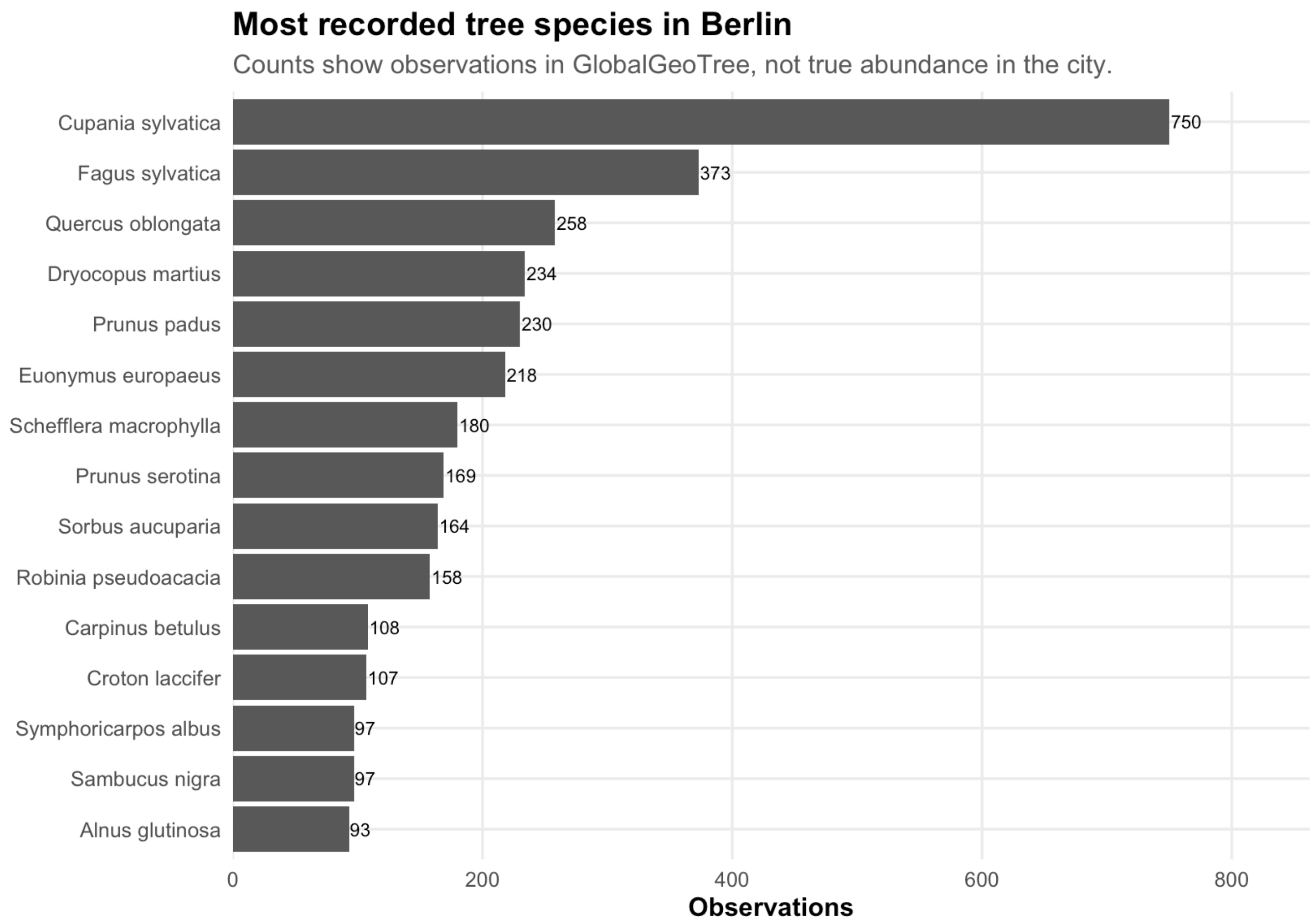
1.6 Top recorded species in Berlin

Why this visual is needed: this gives a simple biological summary after the spatial summaries. It should still be read as recorded observations, not a census of tree abundance.

Top recorded species in Berlin

level3_species	observations	share
Cupania sylvatica	750	14.4%
Fagus sylvatica	373	7.2%
Quercus oblongata	258	5.0%
Dryocopus martius	234	4.5%
Prunus padus	230	4.4%
Euonymus europaeus	218	4.2%

level3_species	observations	share
Schefflera macrophylla	180	3.5%
Prunus serotina	169	3.3%
Sorbus aucuparia	164	3.2%
Robinia pseudoacacia	158	3.0%
Carpinus betulus	108	2.1%
Croton laccifer	107	2.1%
Sambucus nigra	97	1.9%
Symphoricarpos albus	97	1.9%
Alnus glutinosa	93	1.8%



2 Nordrhein-Westfalen (NRW)

2.1 Load NRW boundaries

Why this step is needed: NRW needs one boundary for filtering observations and NUTS3 boundaries for district-level comparisons. NUTS3 regions correspond approximately to Kreise and kreisfreie Städte.

NRW boundary check

item	value
NRW boundary available	Yes
NUTS3 boundary status	Loaded NUTS3 boundaries from data/external/germany_nuts3_2021.geojson
NRW district polygons	53

2.2 Filter observations to NRW

Why this table is needed: before mapping NRW, we need to know how many observations and species were found and whether filtering used real boundaries or a fallback box.

NRW observation summary

metric	value
NRW filtering method	Spatial join to official NUTS1 NRW boundary
NRW observations	36,534
NRW recorded species	243
NRW recorded genera	126
NRW recorded families	61
Available source values	16
Available years	2015-2024

Interpretation. This table is a necessary quality check. If the boundary join is used, the NRW subset is more reliable than a simple bounding-box filter.

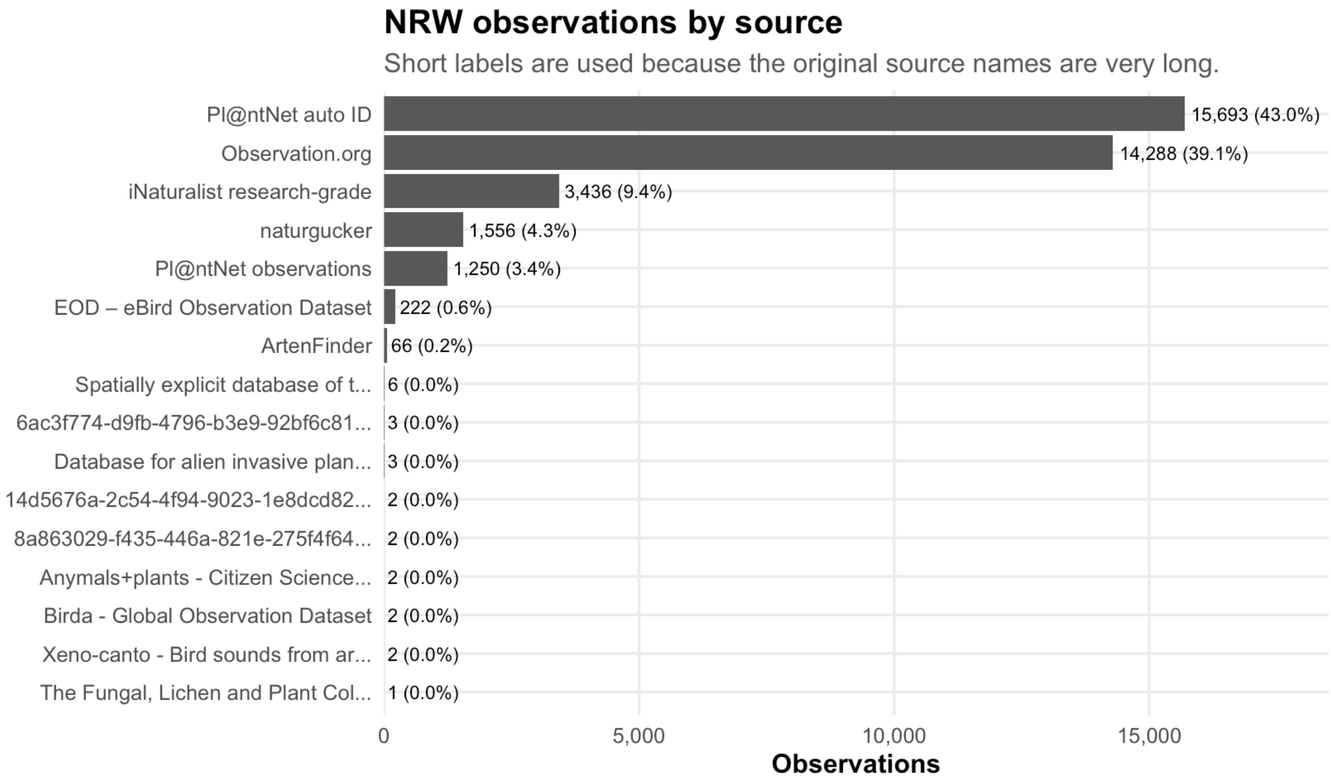
2.3 NRW observation sources

Why this visual is needed: the NRW PDF showed that a few data sources contribute most records, so source composition should stay in the unified report.

NRW observations by source

source_short	observations	share	label
Pl@ntNet auto ID	15,693	43.0%	15,693 (43.0%)
Observation.org	14,288	39.1%	14,288 (39.1%)
iNaturalist research-grade	3,436	9.4%	3,436 (9.4%)
naturgucker	1,556	4.3%	1,556 (4.3%)
Pl@ntNet observations	1,250	3.4%	1,250 (3.4%)
EOD – eBird Observation Dataset	222	0.6%	222 (0.6%)
ArtenFinder	66	0.2%	66 (0.2%)
Spatially explicit database of t…	6	0.0%	6 (0.0%)
6ac3f774-d9fb-4796-b3e9-92bf6c81…	3	0.0%	3 (0.0%)

source_short	observations	share	label
Database for alien invasive plan…	3	0.0%	3 (0.0%)
14d5676a-2c54-4f94-9023-1e8dcd82…	2	0.0%	2 (0.0%)
8a863029-f435-446a-821e-275f4f64…	2	0.0%	2 (0.0%)
Anymals+plants - Citizen Science…	2	0.0%	2 (0.0%)
Birda - Global Observation Dataset	2	0.0%	2 (0.0%)
Xeno-canto - Bird sounds from ar…	2	0.0%	2 (0.0%)
The Fungal, Lichen and Plant Col…	1	0.0%	1 (0.0%)

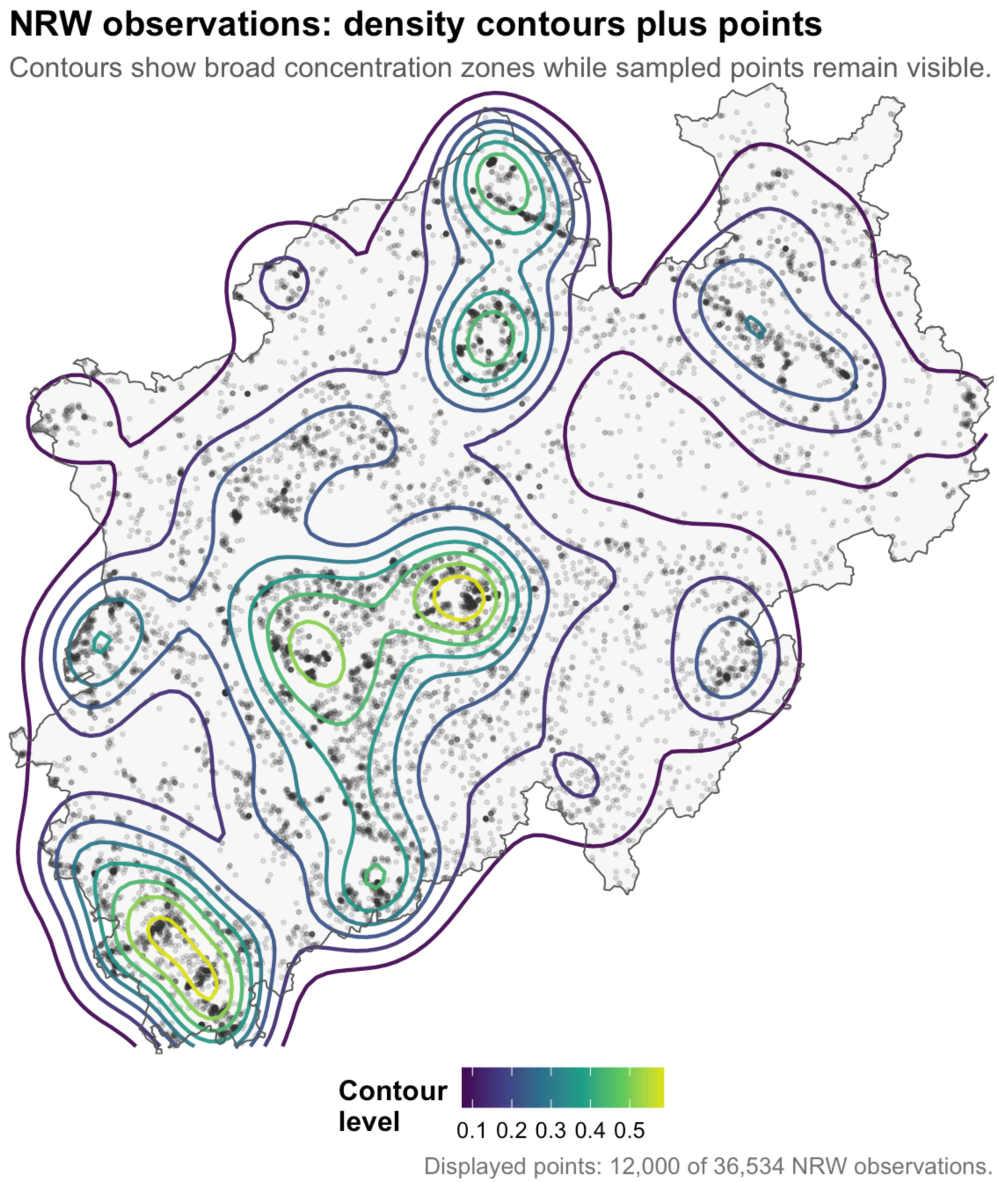


Interpretation. This chart explains source bias before the maps. A dominant source can make one region look more important simply because that platform has more records there.

2.4 NRW observation density maps

Why this visual is needed: a map is needed to see the spatial distribution inside NRW, but a raw point map can be too crowded. Because the best style is not obvious, this report keeps several NRW density visualisations for direct comparison.

My recommendation: the **points + contours** map is the best balanced option. It keeps the underlying point pattern visible, but it also shows smoother hotspot zones. The **smooth filled density** map is also good for presentation.

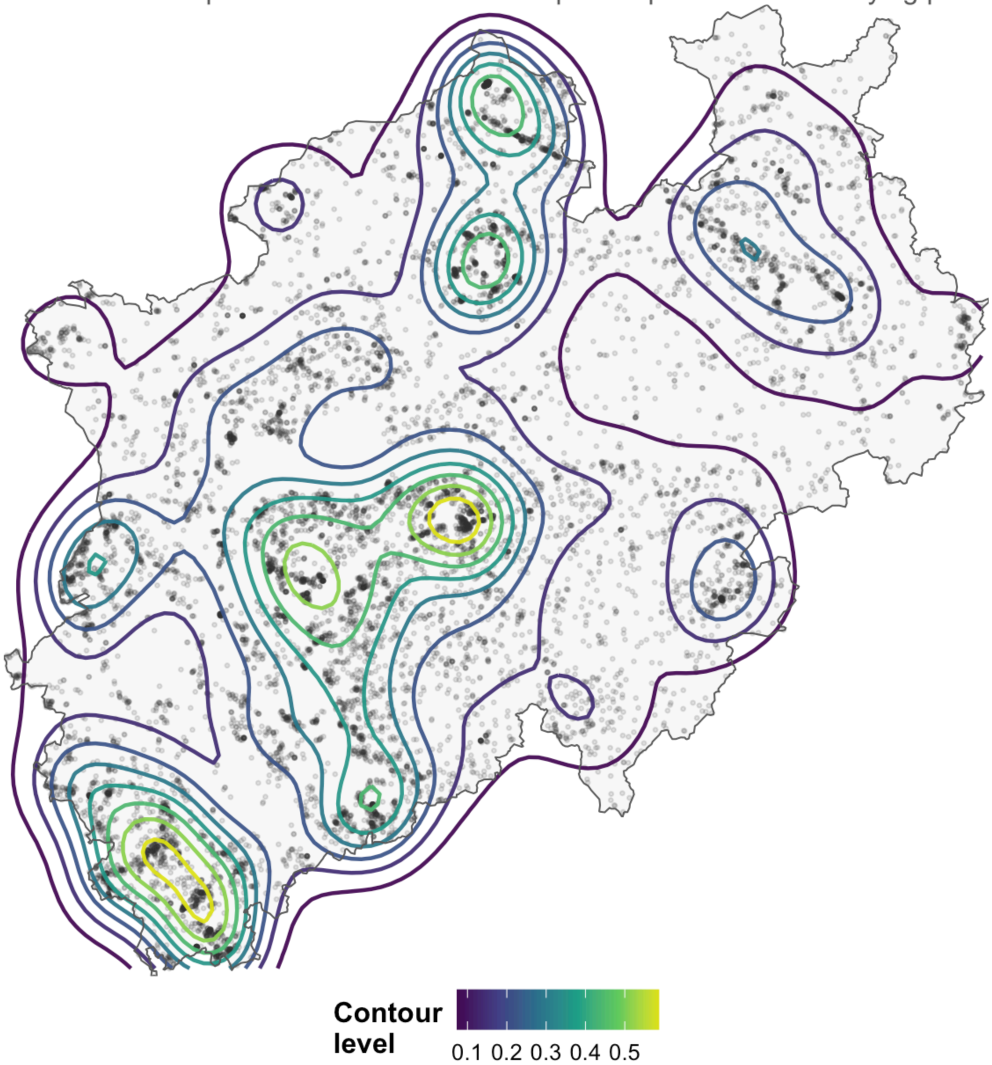


Interpretation. This map is good for seeing NRW hotspots, but it should not be interpreted as true tree abundance. It is a map of where records are concentrated.

2.4.1 NRW density map comparison

NRW observations: density contours plus points

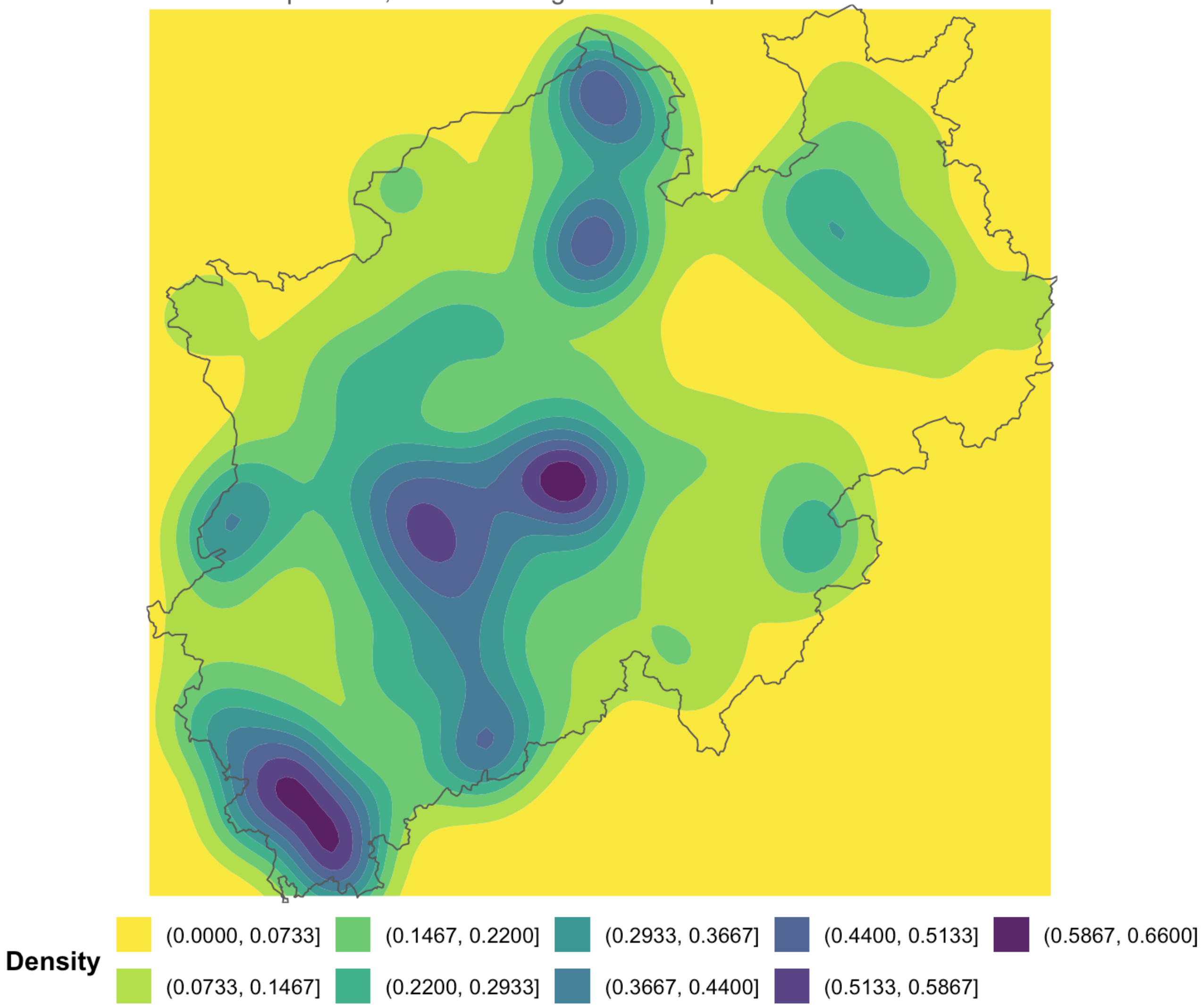
Best balanced option. It shows both the hotspot shape and the underlying point cloud.



Good when you want more structure than a point map, but more honesty than a fully smoothed surface.

NRW observations: smooth filled density

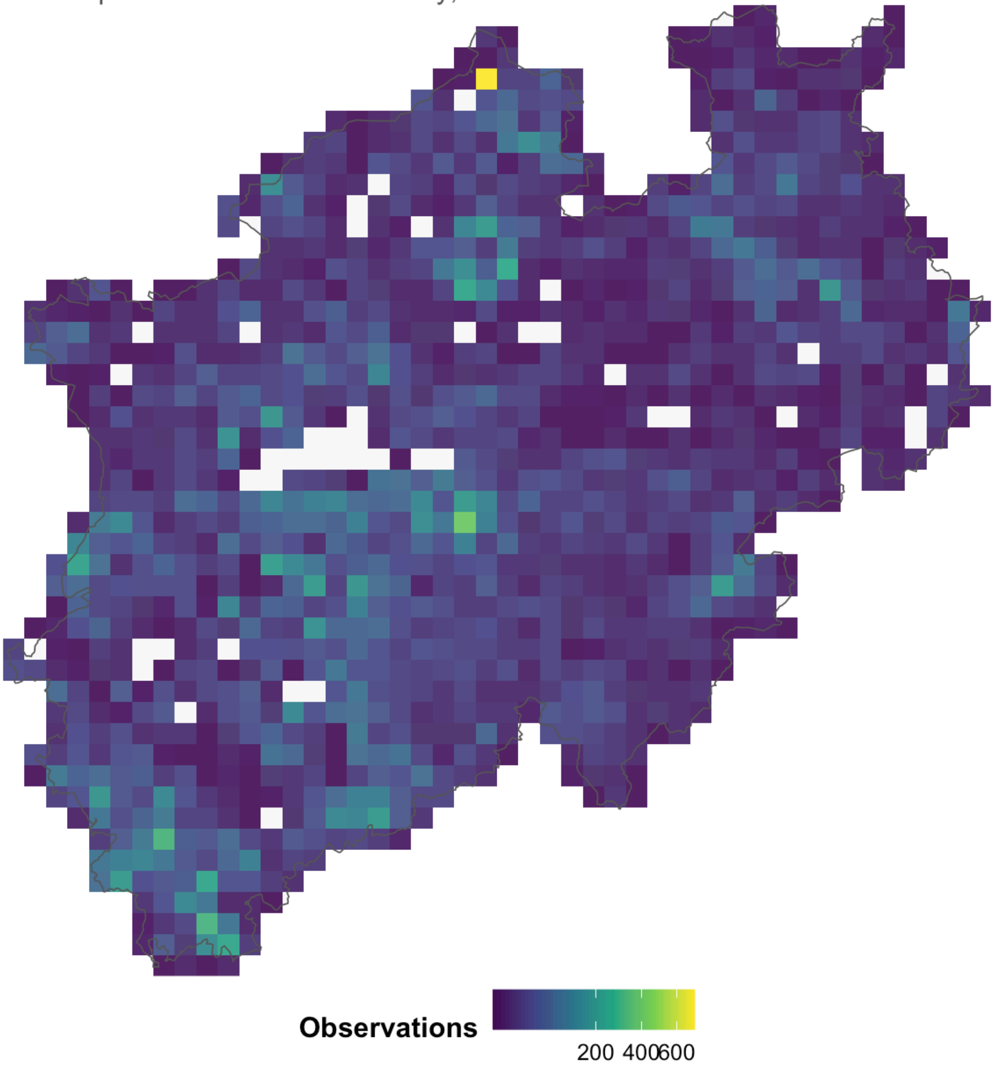
A clear hotspot view, but smoothing can make sparse areas look more continuous than they a



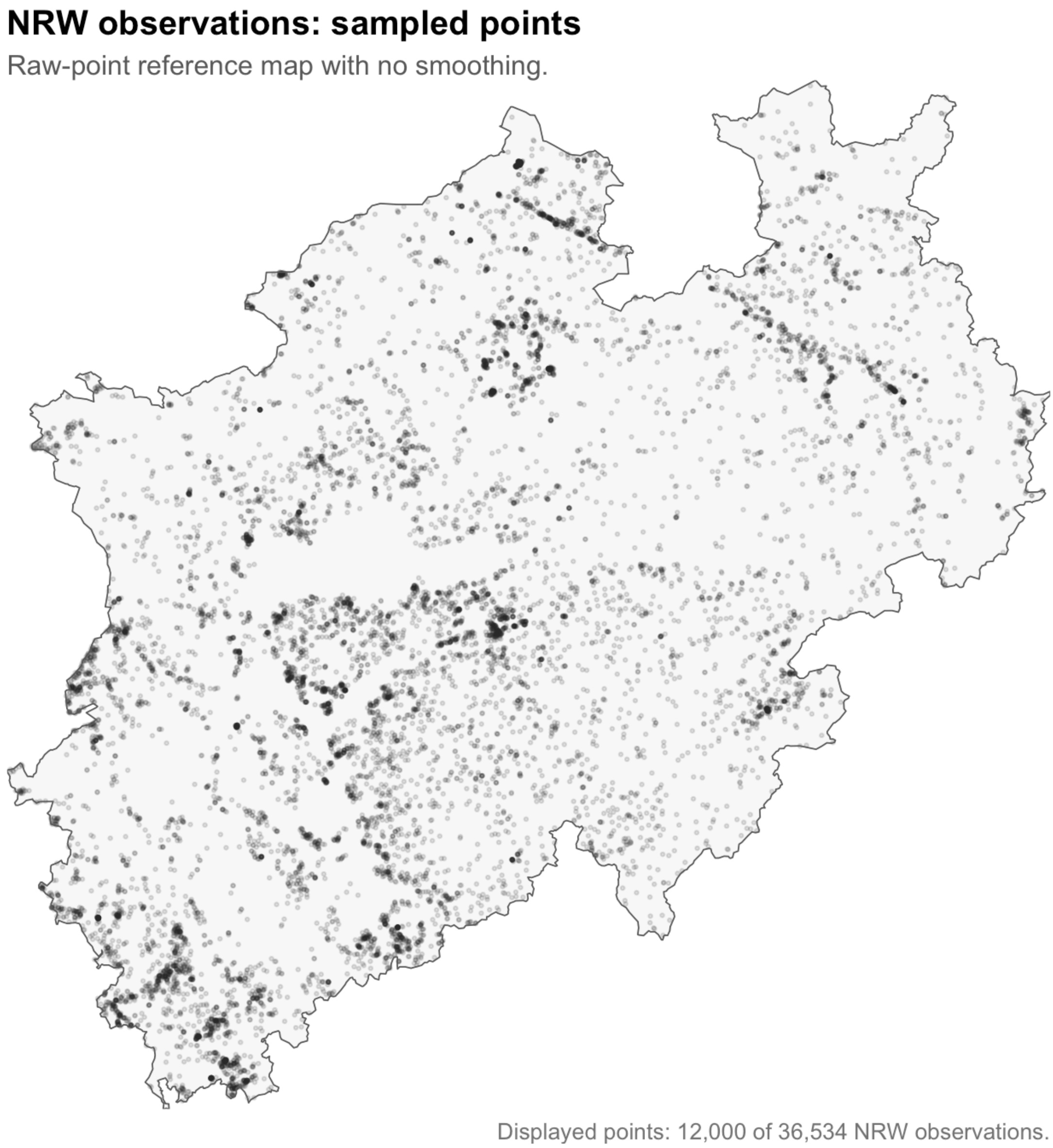
Best for a clean presentation-style hotspot map.

NRW observations: gridded heatmap

Each square counts records directly, so the result is less smoothed and easier to compare cell



Useful for a more count-based spatial summary.



Quick recommendation. If you want to keep two NRW maps, I would keep **density contours plus points** and **smooth filled density**. The contour version is more informative, while the filled version is cleaner.

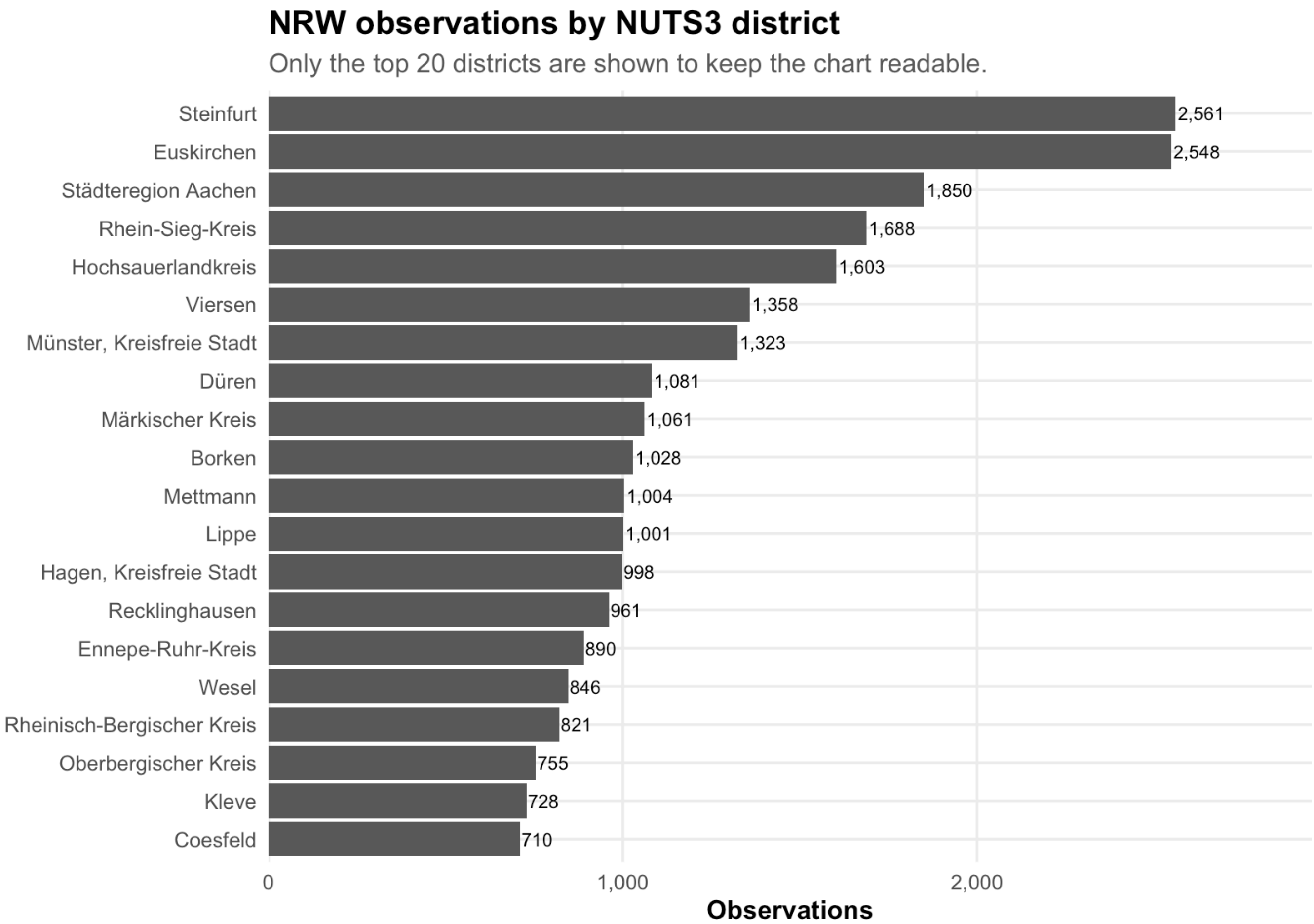
2.5 NRW districts: observations and species

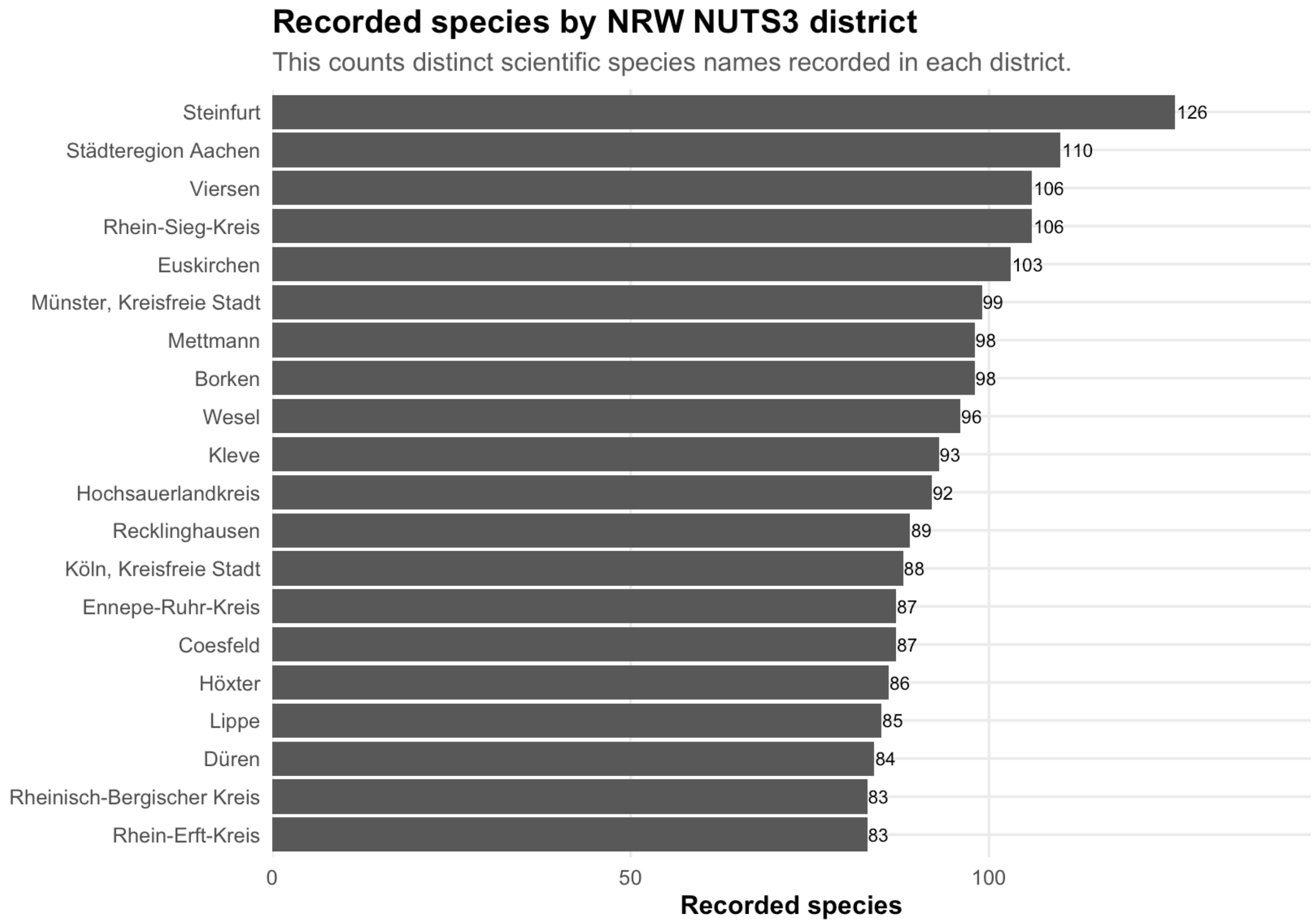
Why this step is needed: NUTS3 districts provide a useful middle scale between individual points and the whole NRW region.

Top NRW NUTS3 districts by observation count

district_id	district_name	observations	species	sources
DEA37	Steinfurt	2,561	126	6
DEA28	Euskirchen	2,548	103	7
DEA2D	Städteregion Aachen	1,850	110	8
DEA2C	Rhein-Sieg-Kreis	1,688	106	7
DEA57	Hochsauerlandkreis	1,603	92	7
DEA1E	Viersen	1,358	106	8

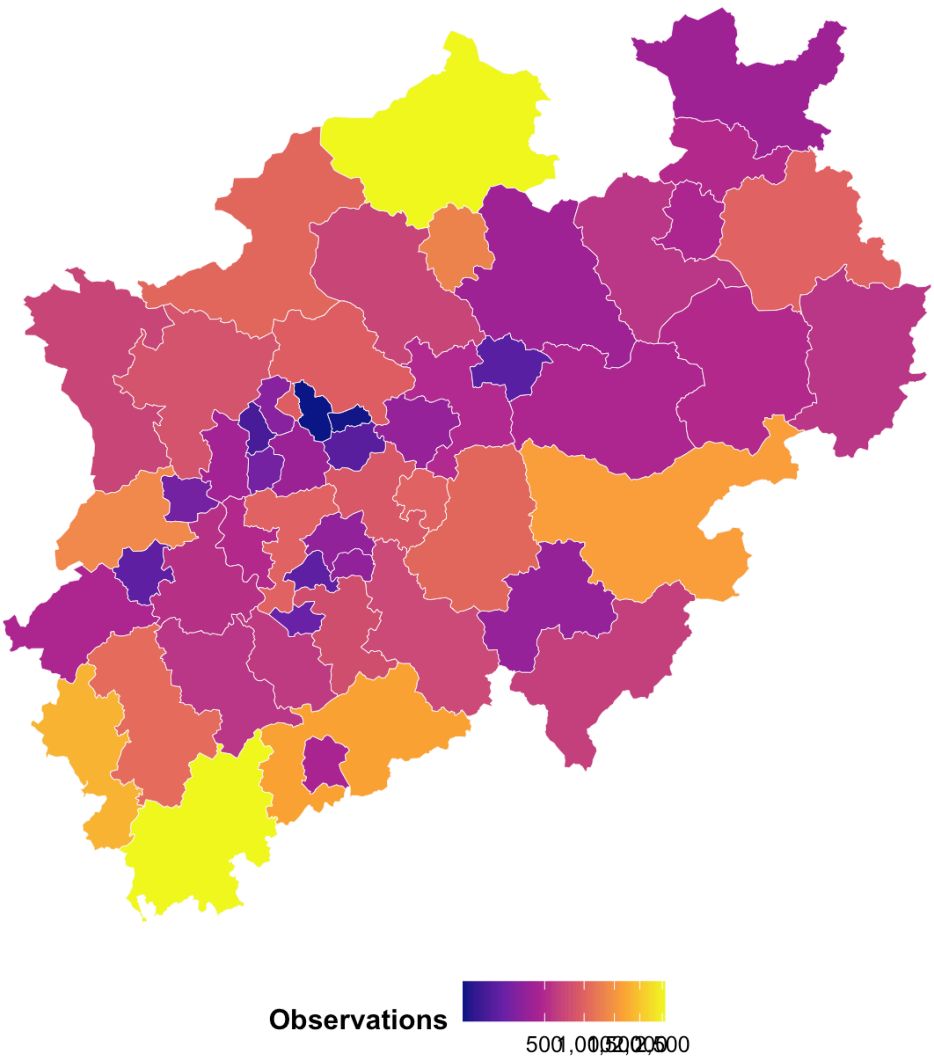
district_id	district_name	observations	species	sources
DEA33	Münster, Kreisfreie Stadt	1,323	99	8
DEA26	Düren	1,081	84	7
DEA58	Märkischer Kreis	1,061	80	7
DEA34	Borken	1,028	98	6
DEA1C	Mettmann	1,004	98	6
DEA45	Lippe	1,001	85	7
DEA53	Hagen, Kreisfreie Stadt	998	79	6
DEA36	Recklinghausen	961	89	7
DEA56	Ennepe-Ruhr-Kreis	890	87	6
DEA1F	Wesel	846	96	6
DEA2B	Rheinisch-Bergischer Kreis	821	83	6
DEA2A	Oberbergischer Kreis	755	74	9
DEA1B	Kleve	728	93	6
DEA35	Coesfeld	710	87	6




Recorded species by NRW NUTS3 district
This counts distinct scientific species names recorded in each district.
Steinfurt 126
Städteregion Aachen 110
Viersen 106
Rhein-Sieg-Kreis 106
Euskirchen 103
Münster, Kreisfreie Stadt 99
Mettmann 98
Borken 98
Wesel 96
Kleve 93
Hochsauerlandkreis 92
Recklinghausen 89
Köln, Kreisfreie Stadt 88
Ennepe-Ruhr-Kreis 87
Coesfeld 87
Höxter 86
Lippe 85
Düren 84
Rheinisch-Bergischer Kreis 83
Rhein-Erft-Kreis 83
0
50
100
Recorded species

Observation frequency across NRW NUTS3 districts

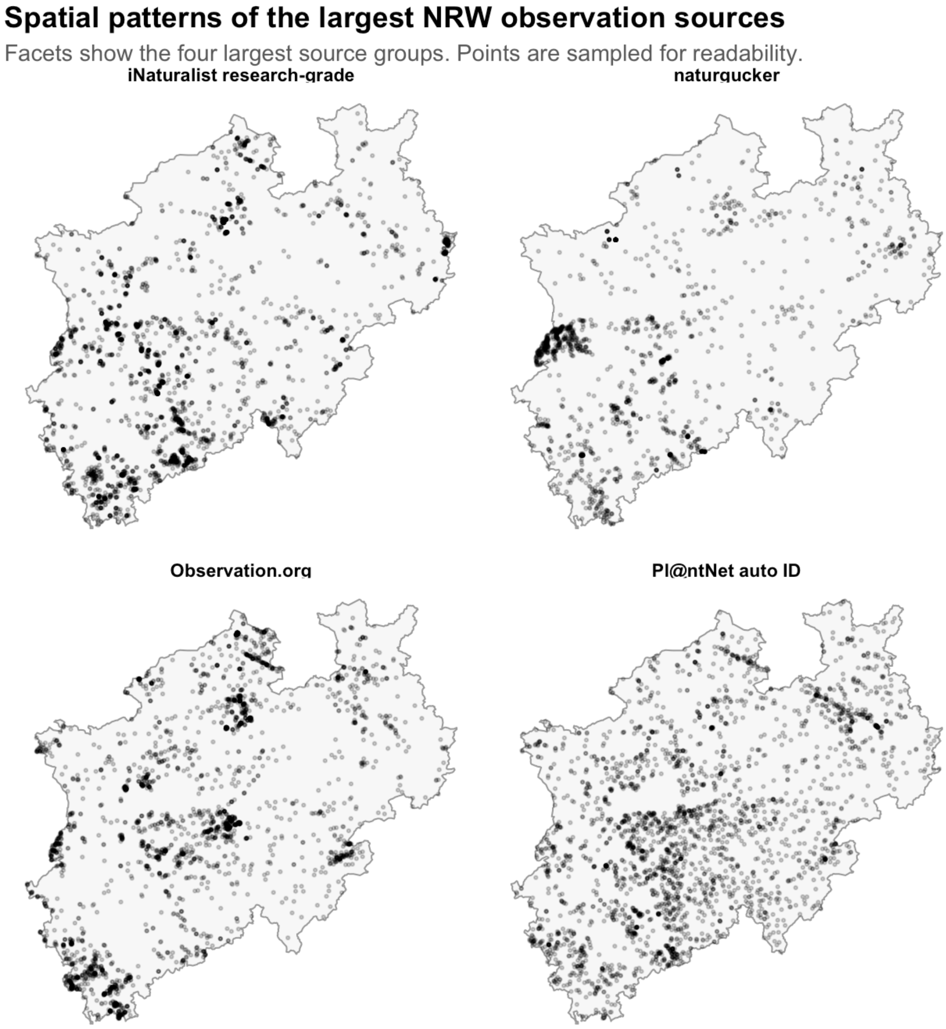
Square-root scaling improves readability when a few districts have very high counts.



Interpretation. The district bar charts are best for ranking. The district map is best for seeing whether high-record districts are clustered geographically.

2.6 Sources on the NRW map

Why this visual is needed: the source bar chart shows size, but this faceted map checks whether the largest sources cover the same parts of NRW.



Interpretation. If source panels look different, NRW spatial patterns may partly reflect source coverage. This makes the source map useful next to the overall density map.

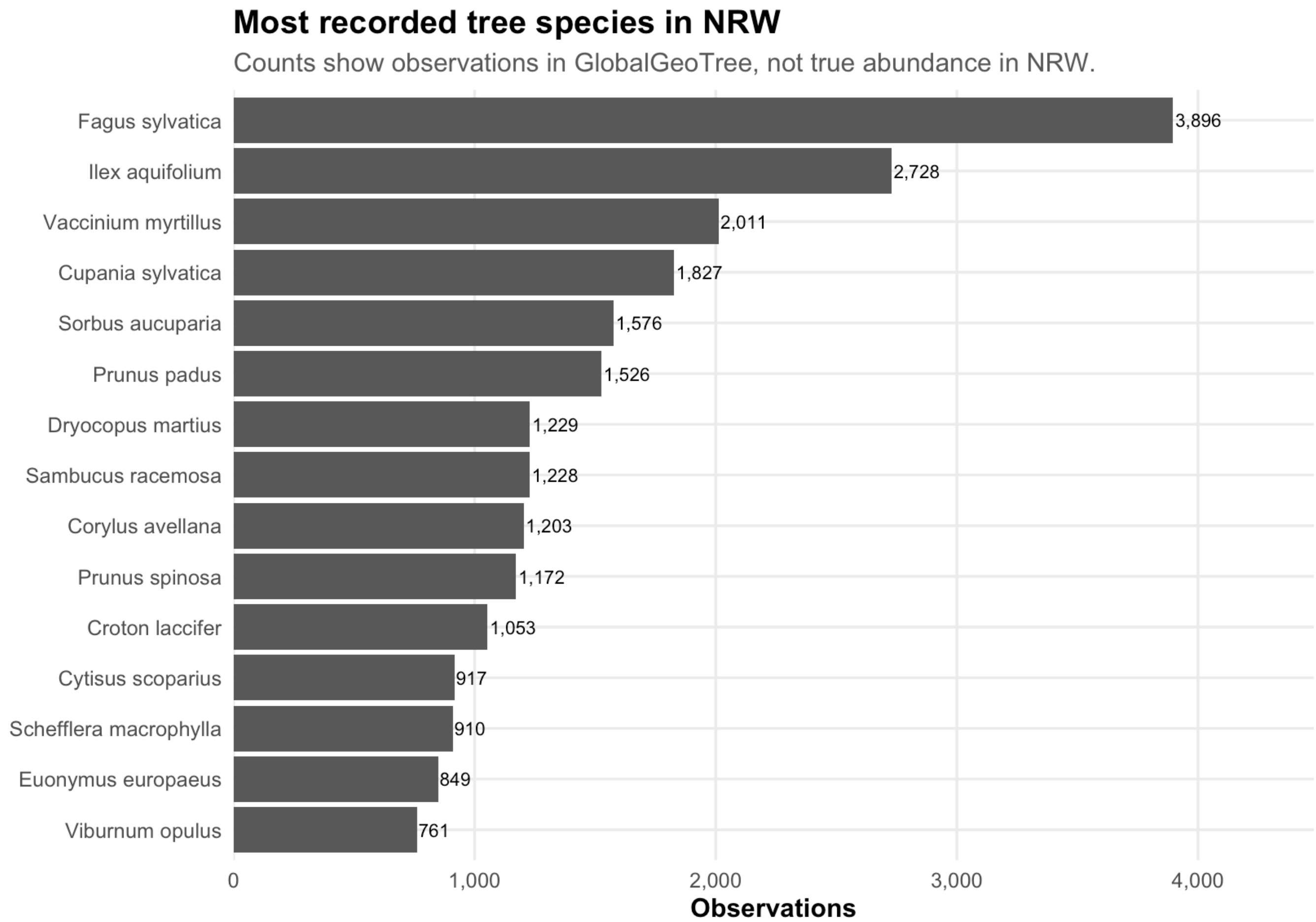
2.7 Top recorded species in NRW

Why this visual is needed: this gives a simple taxonomic summary for NRW without using optional GBIF common-name lookups. Scientific names are kept because they are more reproducible.

Top recorded species in NRW

level3_species	observations	share
Fagus sylvatica	3,896	10.7%
Ilex aquifolium	2,728	7.5%
Vaccinium myrtillus	2,011	5.5%
Cupania sylvatica	1,827	5.0%
Sorbus aucuparia	1,576	4.3%
Prunus padus	1,526	4.2%
Dryocopus martius	1,229	3.4%
Sambucus racemosa	1,228	3.4%
Corylus avellana	1,203	3.3%

level3_species	observations	share
Prunus spinosa	1,172	3.2%
Croton laccifer	1,053	2.9%
Cytisus scoparius	917	2.5%
Schefflera macrophylla	910	2.5%
Euonymus europaeus	849	2.3%
Viburnum opulus	761	2.1%
Quercus oblongata	714	2.0%
Sambucus nigra	605	1.7%
Carpinus betulus	577	1.6%
Alnus glutinosa	462	1.3%
Erica excelsa	440	1.2%



2.8 Final interpretation notes

The most useful visualisations are the ones that compare the same records in different ways: Germany maps for national spatial patterns, year counts for time, source charts for data bias, Bundesländer and NRW district charts for regional comparison, and Berlin density/local-area plots for city-level clustering.

The map and bar-chart pairs are intentionally kept together. Maps are good for spatial patterns, but bar charts are better for exact comparisons.

All maps should be interpreted as maps of **recording intensity** in `GlobalGeoTree.csv`. High values can reflect true tree presence, but they can also reflect where observers and data platforms are more active.